```julia
using CairoMakie
```

Calculating Riemann Sums

In the following exercises, we'll graph each function over the given interval. We'll partition the interval into four subintervals of equal length. Then, we'll add the "rectangles" associated with the Riemann sum

$$\sum_{k=1}^{4} f(c_k)\Delta x,$$

given that c_k is the **(a)** left-hand endpoint, **(b)** right-hand endpoint, **(c)** mindpoint of the `k`th subinterval. (There will be a separate sketch for each set of rectangles.)

Constants:

n = 4

```julia
n = 4 # The number of subintervals
```

Helper functions:

```
riemann_plot (generic function with 1 method)
```

```julia
function riemann_plot(f, a, b, cpoints; title="")
    n = length(cpoints)
    Δ = (b - a) / n
    xs = range(a, b, length=800)

    fig = Figure(size=(900, 500))
    ax = Axis(fig[1, 1], title=title, xlabel="x", ylabel="y")
    hlines!(ax, [0], color=:gray70, linewidth=1)

    lines!(ax, xs, f.(xs), color=:black, linewidth=3)

    for (k, c) in enumerate(cpoints)
        left = a + (k - 1) * Δ
        right = a + k * Δ
        h = f(c)

        poly!(
            ax,
            Point2f[(left, 0), (left, h), (right, h), (right, 0)],
            color=RGBAf(0.2, 0.5, 0.9, 0.22),
            strokecolor=:dodgerblue,
            strokewidth=2
        )

        scatter!(ax, [c], [h], color=:crimson, markersize=12)
        vlines!(ax, [c], color=(:crimson, 0.25), linestyle=:dash)
    end

    fig
end
```

```
partition (generic function with 1 method)
```

```julia
partition(a, Δ) = [a + k * Δ for k in 0:n]
```

```
left_hand_endpoint (generic function with 1 method)
```

```julia
left_hand_endpoint(p) = [p[k] for k in 1:n]
```

```
right_hand_endpoint (generic function with 1 method)
```

```julia
right_hand_endpoint(p) = [p[k] for k in 2:(n + 1)]
```

```
midpoint (generic function with 1 method)
```

```julia
midpoint(p, Δ) = [p[k] + (Δ / 2) for k in 1:n]
```

Ex. 1

$$f(x) = x^2 - 1, \quad [0, 2]$$

```
f1 (generic function with 1 method)
```

```
f1(x) = x^2 - 1
```

```
(0, 2)
```

```
a1, b1 = 0, 2
```

```
Δ1 = 0.5
```

```
Δ1 = (b1 - a1) / n
```

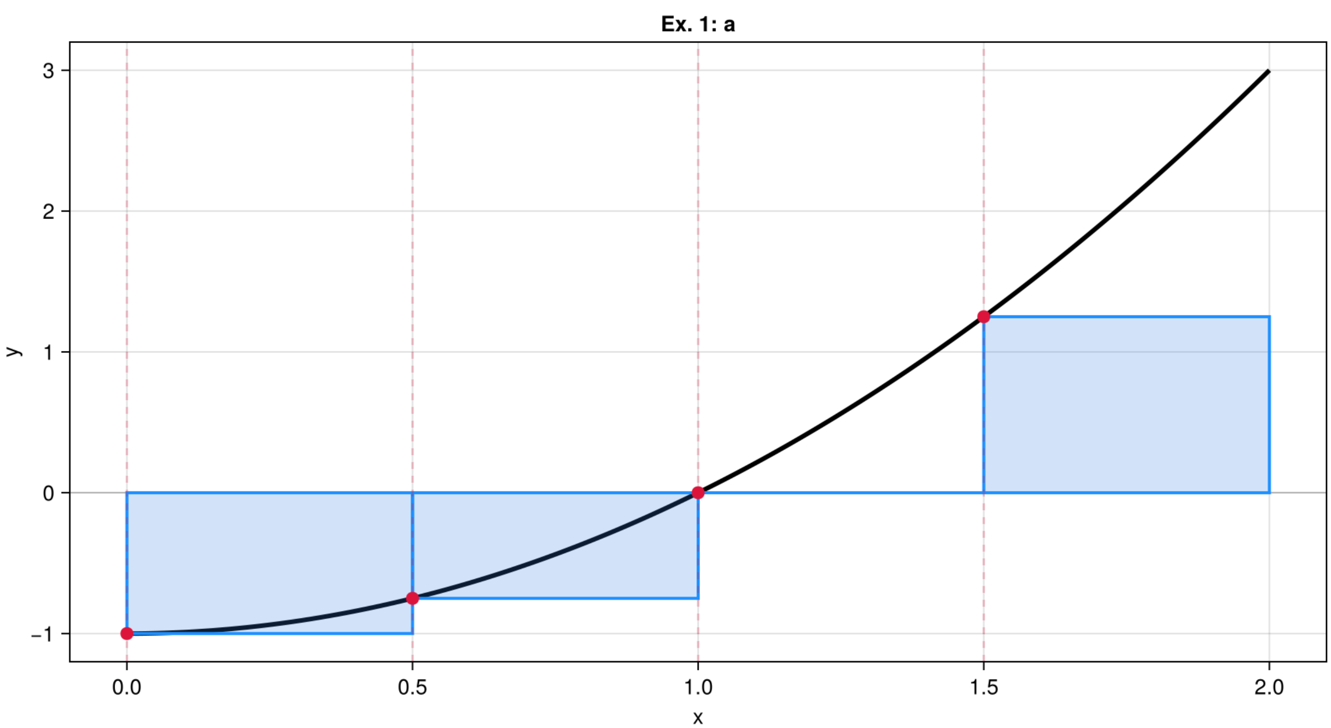
```
partition1 = [0.0, 0.5, 1.0, 1.5, 2.0]
```

```
partition1 = partition(a1, Δ1)
```

(a) Left-hand endpoint:

```
ex1a_cpoints = [0.0, 0.5, 1.0, 1.5]
```

```
ex1a_cpoints = left_hand_endpoint(partition1)
```

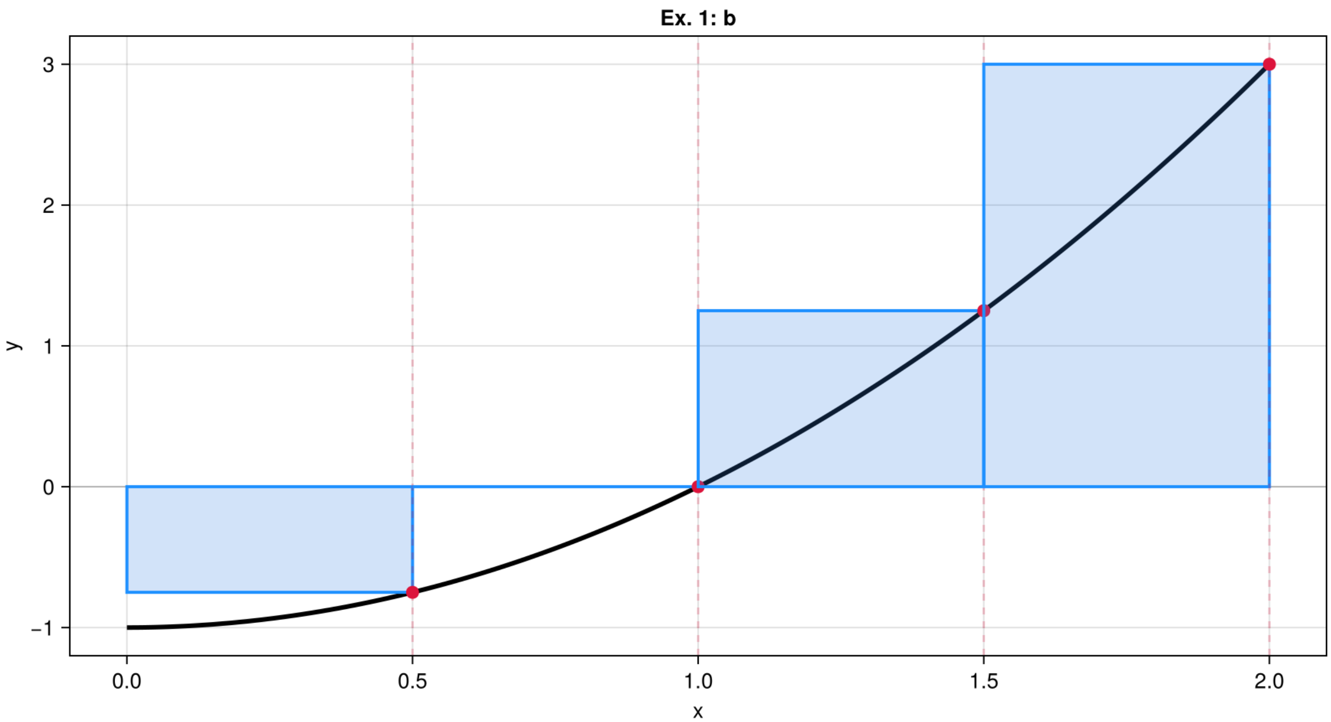



```
riemann_plot(f1, a1, b1, ex1a_cpoints; title="Ex. 1: a")
```

(b) Right-hand endpoint:

```
ex1b_cpoints = [0.5, 1.0, 1.5, 2.0]
```

```
ex1b_cpoints = right_hand_endpoint(partition1)
```

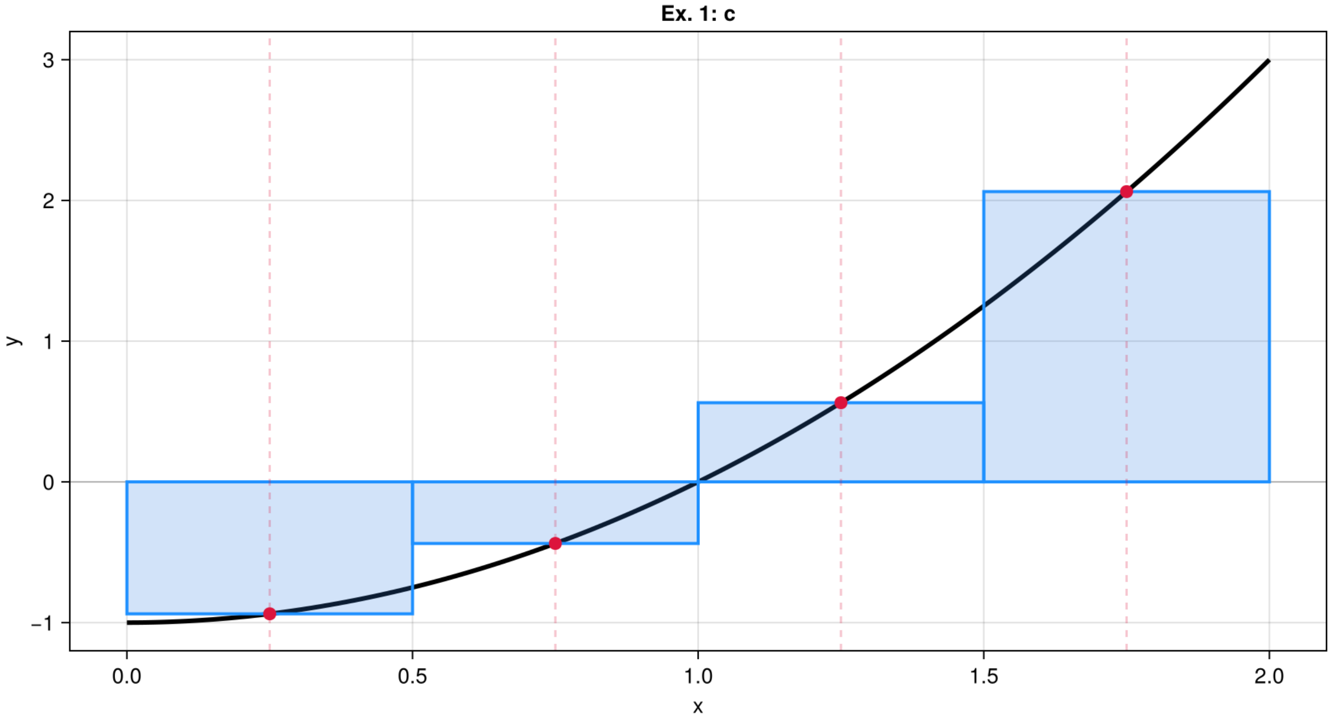



```
1 riemann_plot(f1, a1, b1, ex1b_cpoints; title="Ex. 1: b")
```

(c) Midpoint of the k th subinterval:

ex1c_cpoints = [0.25, 0.75, 1.25, 1.75]

```
1 ex1c_cpoints = midpoint(partition1, Δ1)
```




```
riemann_plot(f1, a1, b1, ex1c_cpoints; title="Ex. 1: c")
```

Ex. 2

$$f(x) = -x^2, \quad [0, 1]$$

f2 (generic function with 1 method)

```
f2(x) = -x^2
```

(0, 1)

```
a2, b2 = 0, 1
```

Δ2 = 0.25

```
Δ2 = (b2 - a2) / n
```

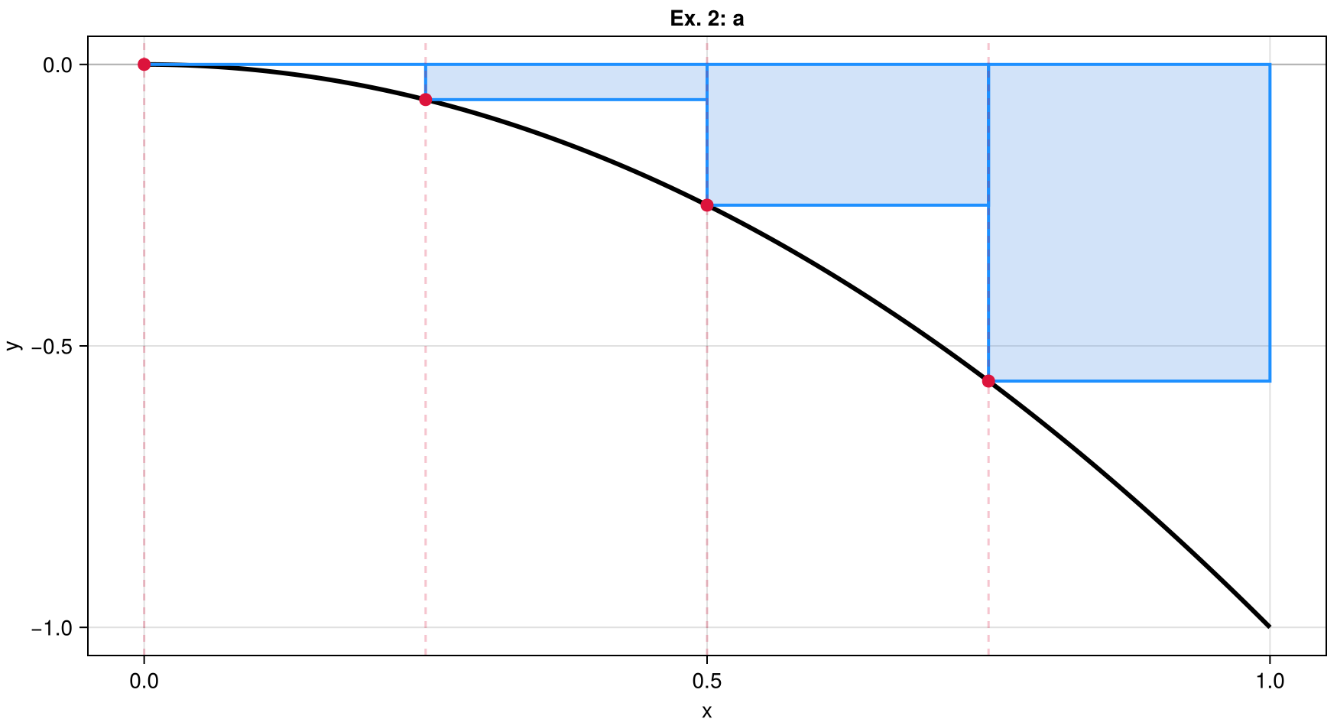
partition2 = [0.0, 0.25, 0.5, 0.75, 1.0]

```
partition2 = partition(a2, Δ2)
```

(a) Left-hand endpoint:

ex2a_cpoints = [0.0, 0.25, 0.5, 0.75]

```
ex2a_cpoints = left_hand_endpoint(partition2)
```

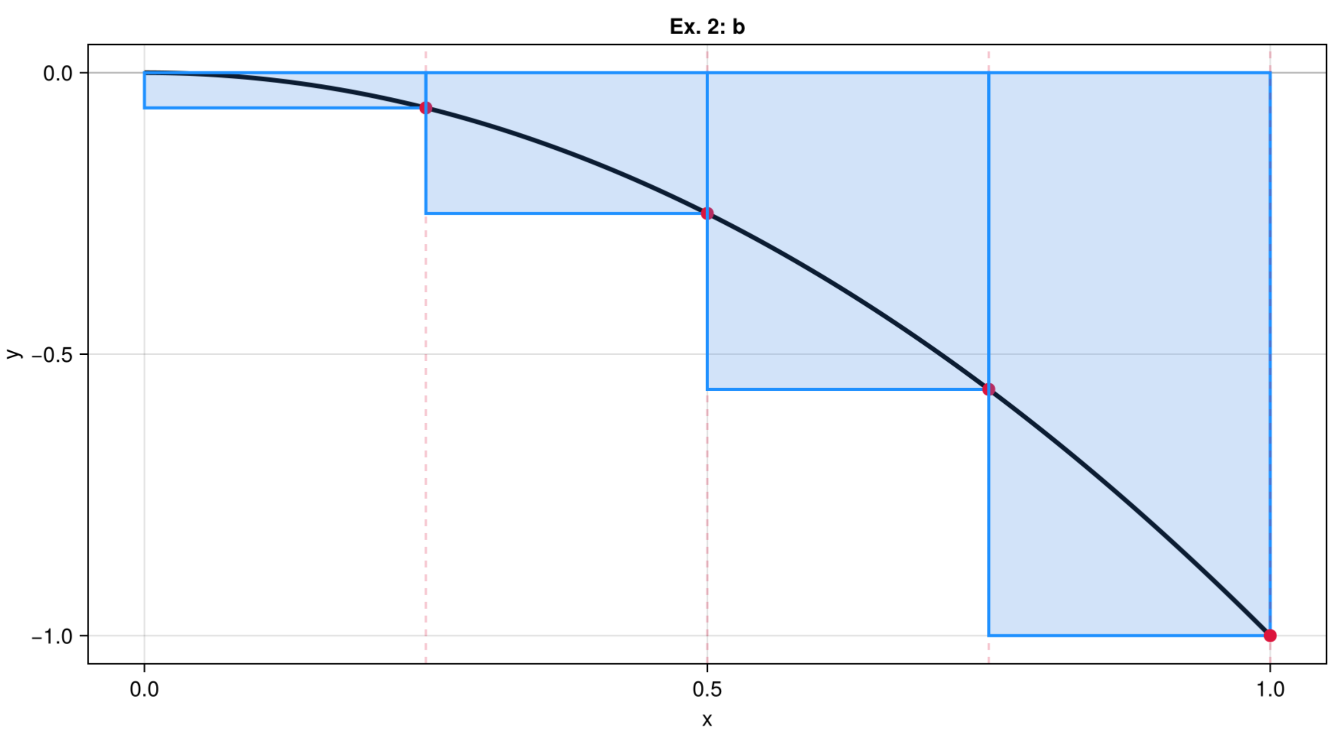



```
riemann_plot(f2, a2, b2, ex2a_cpoints; title="Ex. 2: a")
```

(b) Right-hand endpoint:

```
ex2b_cpoints = [0.25, 0.5, 0.75, 1.0]
```

```
ex2b_cpoints = right_hand_endpoint(partition2)
```

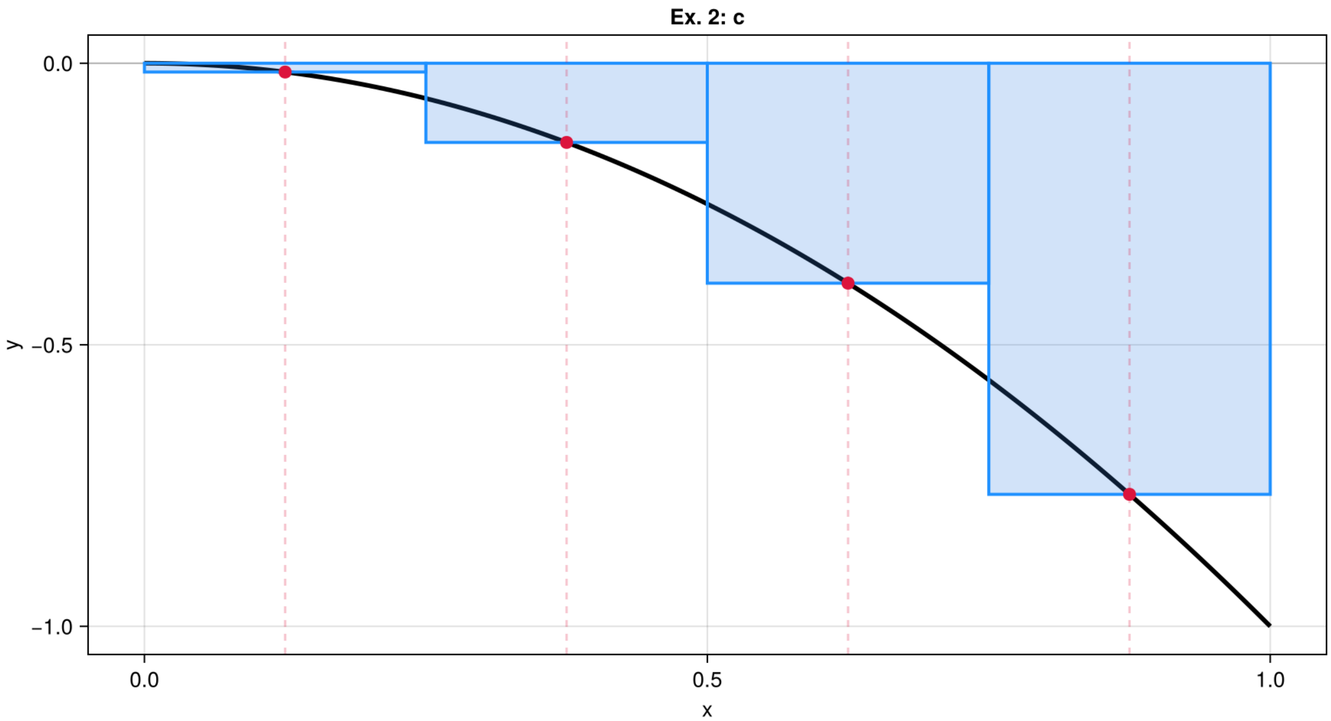



```
riemann_plot(f2, a2, b2, ex2b_cpoints; title="Ex. 2: b")
```

(c) Midpoint of the k th subinterval:

ex2c_cpoints = [0.125, 0.375, 0.625, 0.875]

```
ex2c_cpoints = midpoint(partition2, Δ2)
```




```
riemann_plot(f2, a2, b2, ex2c_cpoints; title="Ex. 2: c")
```

Ex. 3

$$f(x) = sin(x), \quad [-\pi, \pi]$$

f3 (generic function with 1 method)

```
f3(x) = sin(x)
```

(-3.14159, π)

```
a3, b3 = -π, π
```

Δ3 = 1.5707963267948966

```
Δ3 = (b3 - a3) / n
```

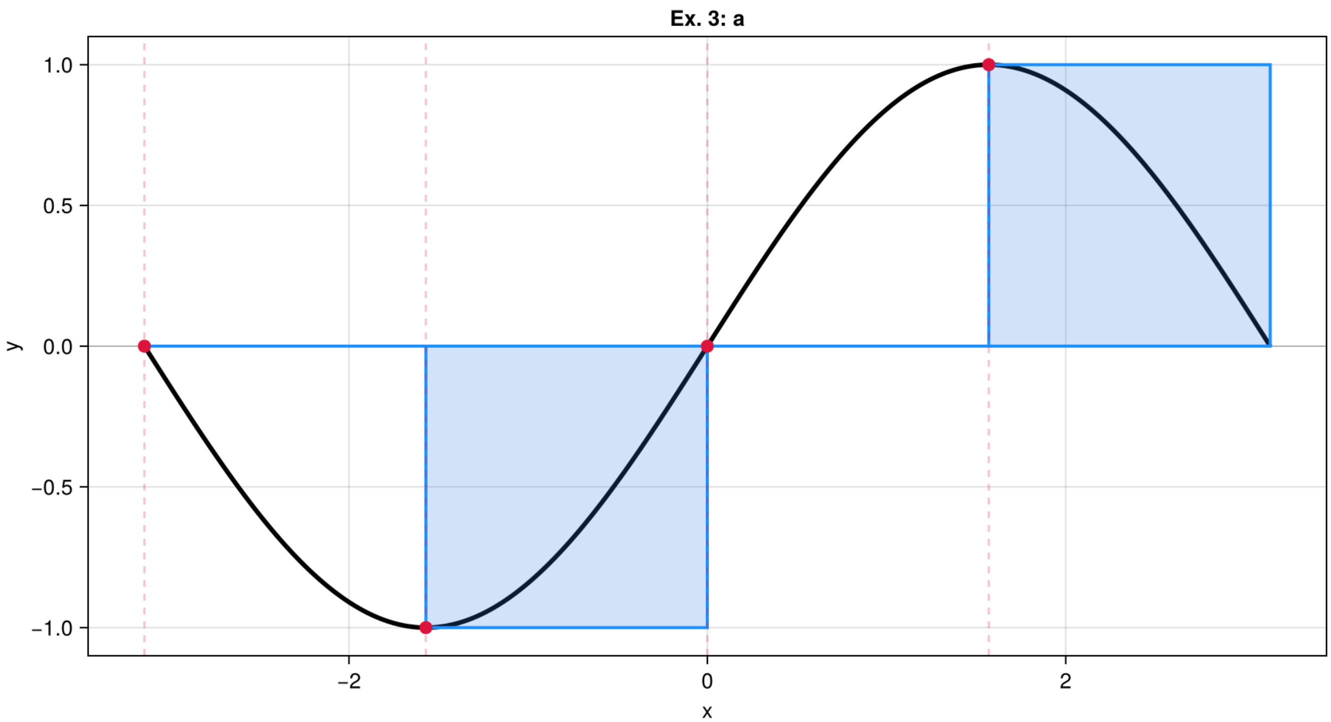
partition3 = [-3.14159, -1.5708, 0.0, 1.5708, 3.14159]

```
partition3 = partition(a3, Δ3)
```

(a) Left-hand endpoint:

```
ex3a_cpoints =  [-3.14159, -1.5708, 0.0, 1.5708]
```

```
ex3a_cpoints = left_hand_endpoint(partition3)
```

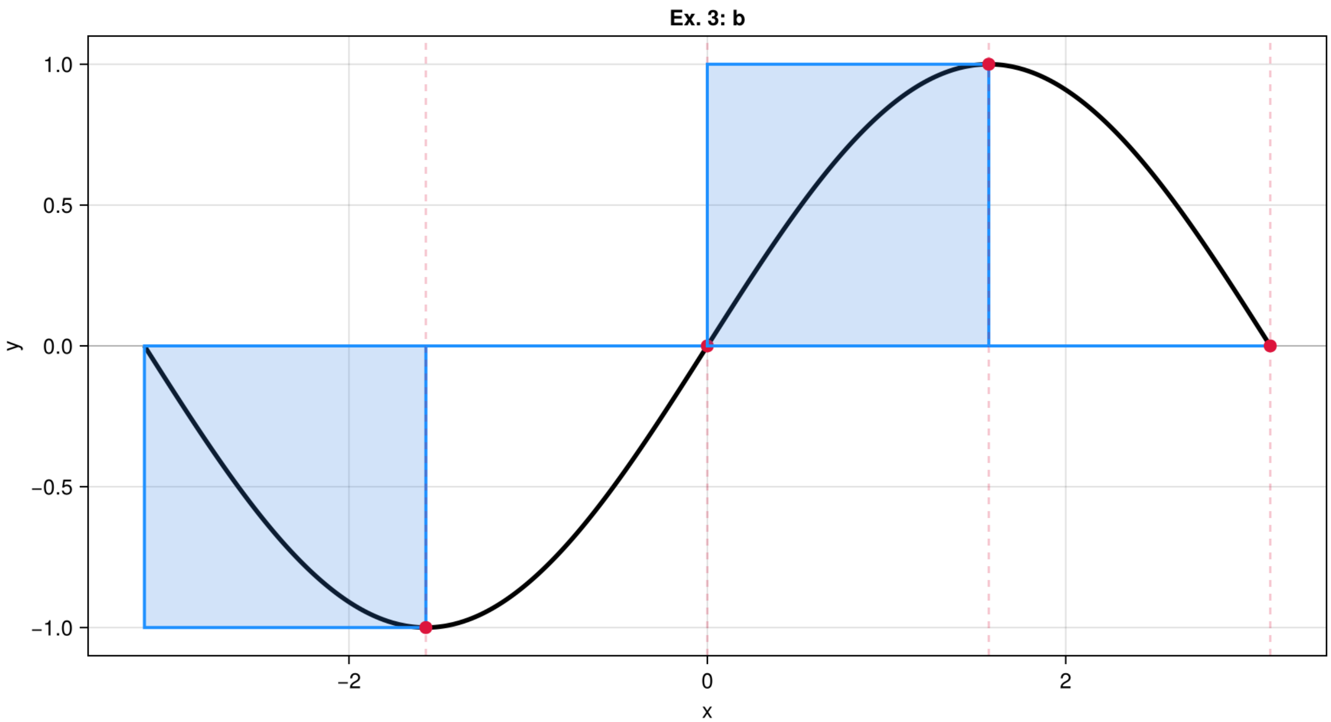



```
riemann_plot(f3, a3, b3, ex3a_cpoints; title="Ex. 3: a")
```

(b) Right-hand endpoint:

```
ex3b_cpoints =  [-1.5708, 0.0, 1.5708, 3.14159]
```

```
ex3b_cpoints = right_hand_endpoint(partition3)
```

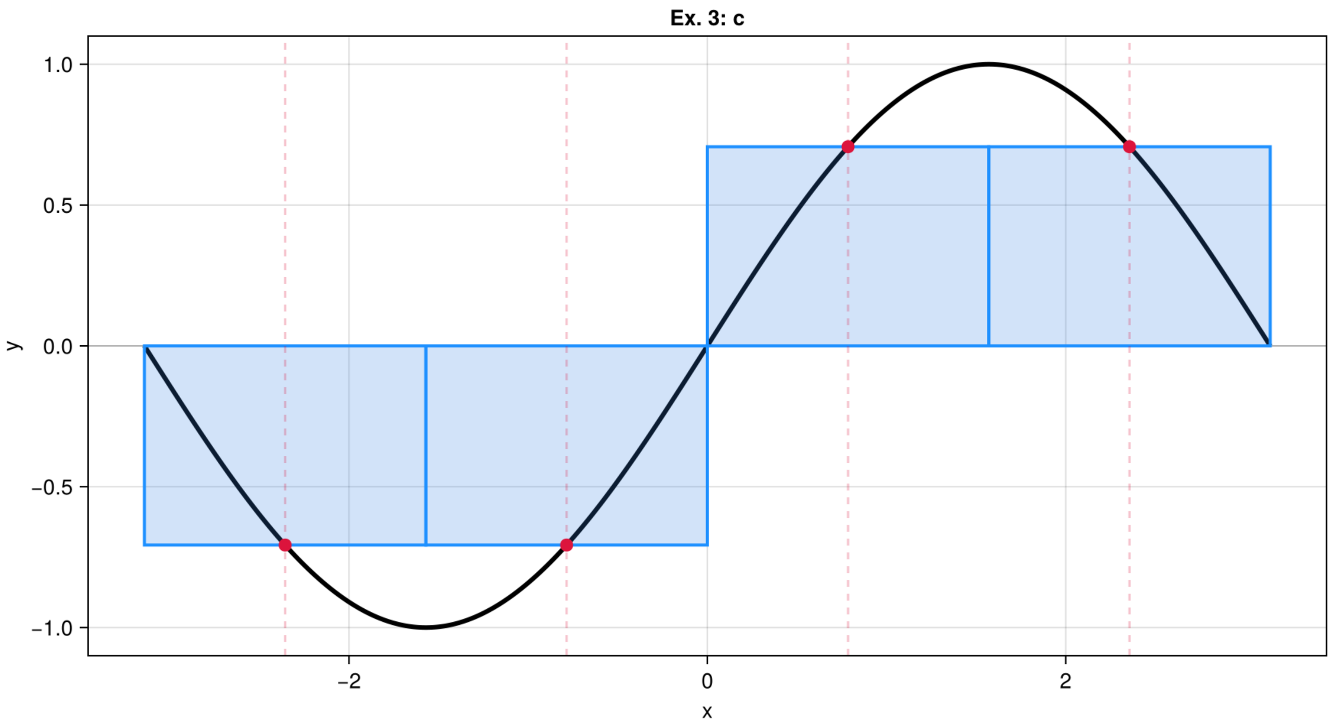



```
1 riemann_plot(f3, a3, b3, ex3b_cpoints; title="Ex. 3: b")
```

(c) Midpoint of the k th subinterval:

```
ex3c_cpoints = [-2.35619, -0.785398, 0.785398, 2.35619]
```

```
1 ex3c_cpoints = midpoint(partition3, Δ3)
```




```
1 riemann_plot(f3, a3, b3, ex3c_cpoints; title="Ex. 3: c")
```

Ex. 4

$$f(x) = sin(x) + 1, \quad [-\pi, \pi]$$

f4 (generic function with 1 method)

```
1 f4(x) = sin(x) + 1
```

(-3.14159, π)

```
1 a4, b4 = -π, π
```

Δ4 = 1.5707963267948966

```
1 Δ4 = (b4 - a4) / n
```

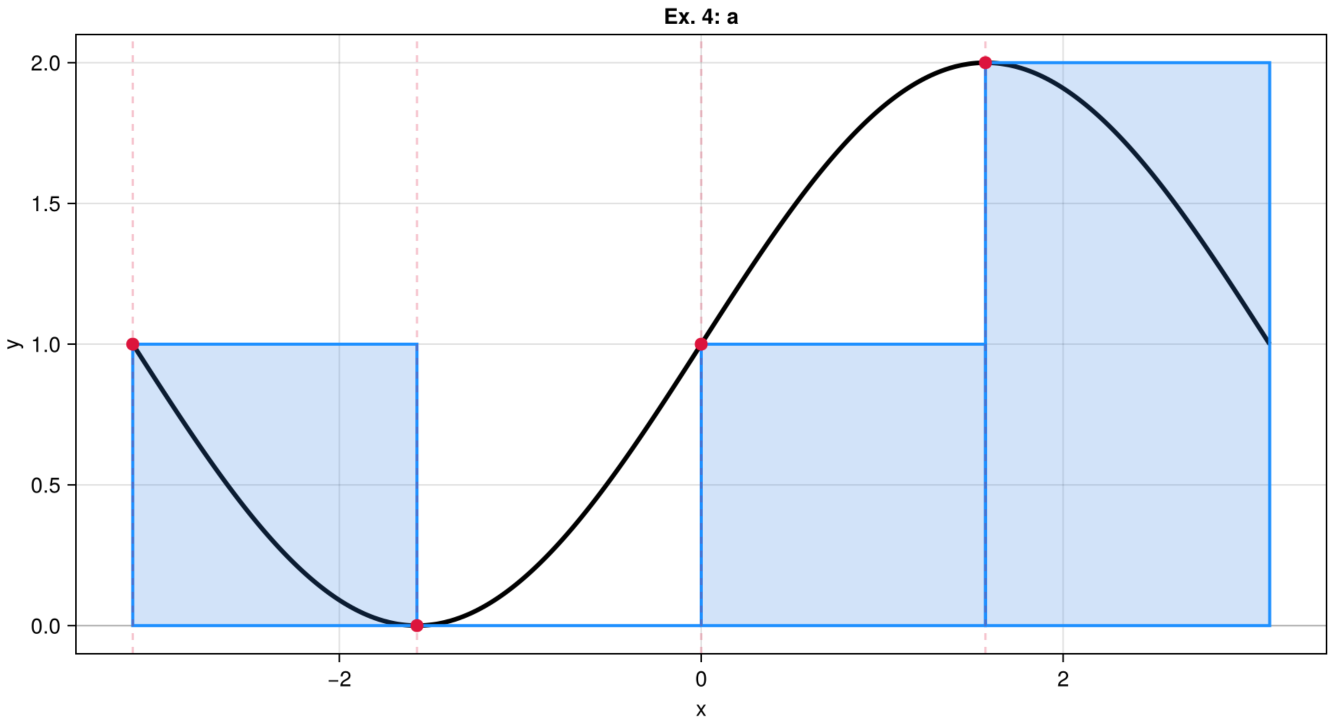
partition4 = [-3.14159, -1.5708, 0.0, 1.5708, 3.14159]

```
1 partition4 = partition(a4, Δ4)
```

(a) Left-hand endpoint:

ex4a_cpoints = [-3.14159, -1.5708, 0.0, 1.5708]

```
1 ex4a_cpoints = left_hand_endpoint(partition4)
```

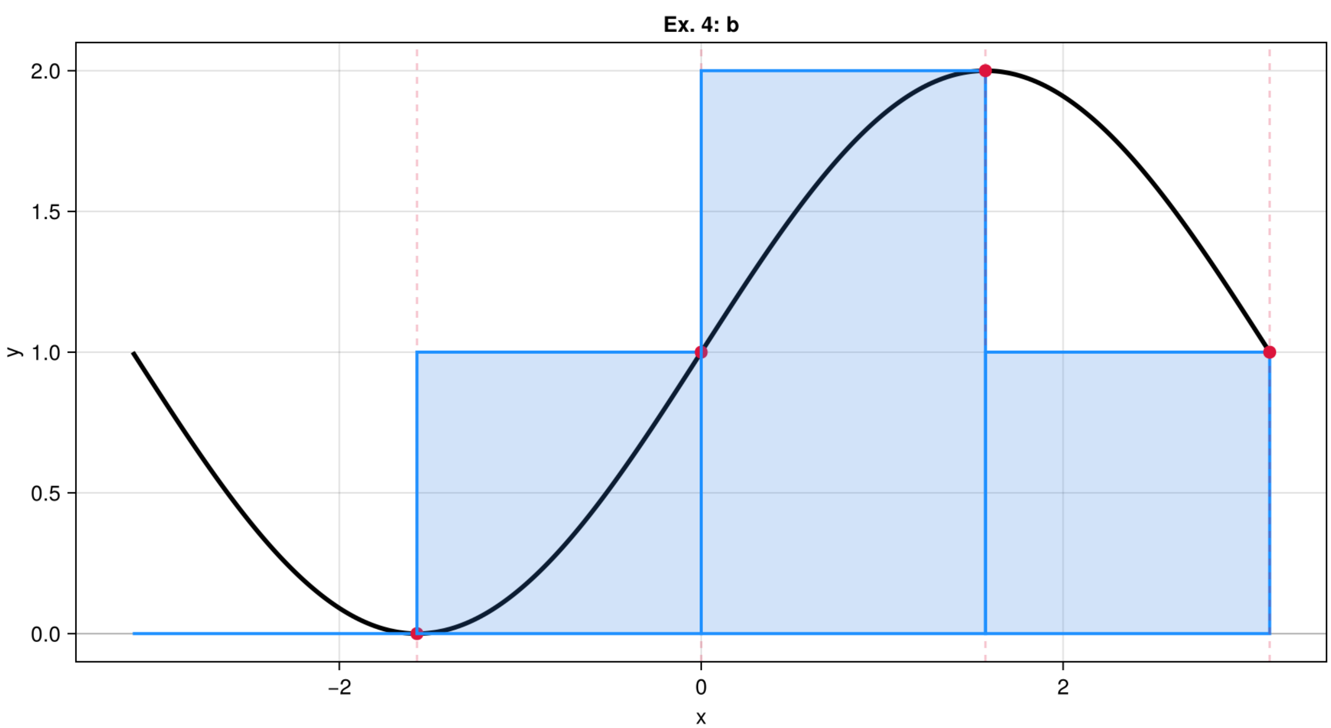



```
riemann_plot(f4, a4, b4, ex4a_cpoints; title="Ex. 4: a")
```

(b) Right-hand endpoint:

ex4b_cpoints = [-1.5708, 0.0, 1.5708, 3.14159]

```
ex4b_cpoints = right_hand_endpoint(partition4)
```

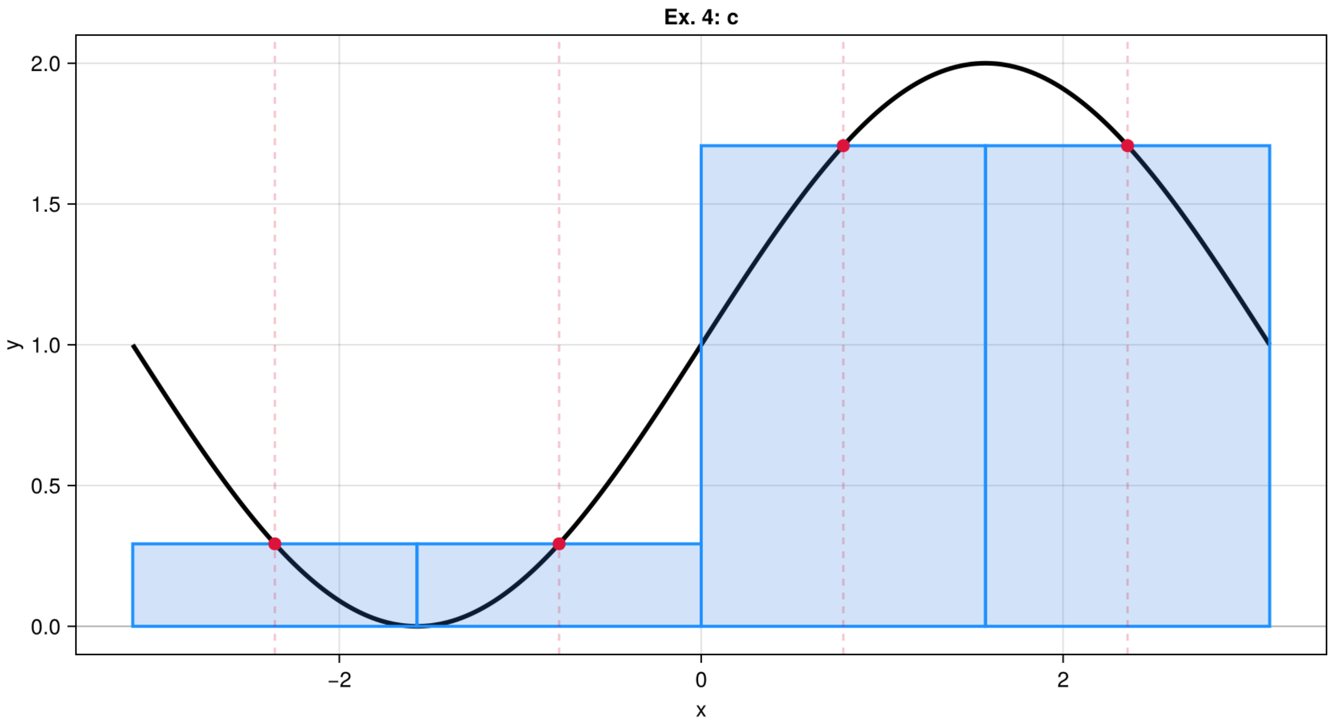



```
riemann_plot(f4, a4, b4, ex4b_cpoints; title="Ex. 4: b")
```

(c) Midpoint of the k th subinterval:

ex4c_cpoints = [-2.35619, -0.785398, 0.785398, 2.35619]

```
ex4c_cpoints = midpoint(partition4, Δ4)
```




```
riemann_plot(f4, a4, b4, ex4c_cpoints; title="Ex. 4: c")
```

Finding the norm of a partition

We define the **norm** of a partition P, written $||P||$, to be the largest of all the subinterval widths.

Ex. 1

$$P = \{0, 1.2, 1.5, 2.3, 2.6, 3\}.$$

P1 = [0.0, 1.2, 1.5, 2.3, 2.6, 3.0]

```
P1 = [0, 1.2, 1.5, 2.3, 2.6, 3]
```

P1_widths = [1.2, 0.3, 0.8, 0.3, 0.4]

```
P1_widths = diff(P1)
```

P1_norm = 1.2

```
P1_norm = maximum(P1_widths)
```

Ex. 2

$$P = \{-2, -1.6, -0.5, 0, 0.8, 1\}.$$

P2 = [-2.0, -1.6, -0.5, 0.0, 0.8, 1.0]

```
P2 = [-2, -1.6, -0.5, 0, 0.8, 1]
```

P2_widths = [0.4, 1.1, 0.5, 0.8, 0.2]

```
P2_widths = diff(P2)
```

P2_norm = 1.1

```
P2_norm = maximum(P2_widths)
```